Harsh Oza

Software Engineer, Samsung R&D Institute India – Bangalore · B.Tech. in CSE, IIT Kanpur
2003harshoza@gmail.com · +91-8735005656

Education

Year	Degree / Examination	Institution / Board	CPI / Percentage
2021 – 2025	B.Tech., Computer Science & Engg.	Indian Institute of Technology Kanpur	8.0 / 10
2021	Class XII (GSEB)	Eklavya School of Science, Ahmedabad	93.0%
2019	Class X (CBSE)	Kendriya Vidyalaya Shahibaug, Ahmedabad	96.0%

Academic Minor: Minor in Philosophy, Department of Humanities and Social Sciences, IIT Kanpur.

Awards and Achievements

- **SPOT Award**, Samsung R&D Institute India – Bangalore: Core contributions to agent memory architecture. *2026*
- Became **Workera AI Expert+** at Samsung, demonstrating proficiency across ML, DL, Vision, and RAG. *2026*
- Cleared Samsung Software Competency (**SWC**) Advanced & Pro tests for DSA problem solving; awarded PPO. *2025*
- Secured an **All India Rank** of **754** in Joint Entrance Examination (**JEE Adv 2021**) out of 1.5 lakh students. *2021*
- Secured an **All-India Rank** of **672** in Joint Entrance Examination (**JEE Main 2021**) out of 10 lakh students. *2021*
- Took part in Indian Olympiad Qualifier in Physics-2020-21 (**IOQP**) and in **top 1411** out of 2 lakh students. *2021*

Research and Professional Experience

On-Device Agent Memory Subsystem (Tizen AIOS) *Jul 2025 – Present*
Software Engineer (AI/ML), Samsung R&D Institute India – Bangalore | Part Head: Raveendra Karu

- Engineered long-term memory engine for on-device AI agents on Samsung Smart TVs under strict RAM and storage limits; re-architected storage from flat Markdown into structured **SQLite** with stable UUIDs, bullet-provenance (`dream_bullet_sources`), and atomic commits with rollback.
- Extended LLM memory consolidation ("dreaming") pipeline to 6 verdicts (adding *update* and *delete* primitives for fact correction); built an atomic **Forget directive** pipeline enforcing sanitized user deletion requests with read/write validation guards.
- Built a **Rust** benchmarking framework evaluating 6 memory backends on **LoCoMo** conversational dataset (∼90% accuracy vs. 8.5% keyword; HNSW maintained 8–9 ms query latency at 10K scale); evaluated 5 embedding models and 11 LLMs, applying **INT8 quantization** (75% size reduction).

Efficient Text-to-Video Retrieval via Temporal Attention *May 2025 – Present*
Research Project, Samsung Research | Efficient Multimodal Representation Learning

- Formulated **STAdapter V2+**, replacing 1D temporal convolution with a local-window multi-head self-attention module spanning a ±6-frame temporal context window with content-dependent aggregation to adapt static vision-language models for video retrieval.
- Achieved a +9.5% **gain in R_sum@1** on MSR-VTT adding only **16.7K parameters** (+9.2%) at 1.07 ms inference latency; demonstrated compact **MobileCLIP-S0** (54M) consistently outperformed **CLIP ViT-B/32** (151M) across 12 metrics. Authored 1,660-line technical audit.

On-Device Edge Video Analytics & Gesture Recognition *Jul 2025 – Sep 2025*
Software Engineer, SmartThings Edge Framework, Samsung R&D Institute India – Bangalore

- Built a real-time multimodal edge analytics pipeline in C++ for Android using **NNStreamer** and **GStreamer**, concurrently executing **ResNet18** (gesture classification, 34 classes) and **YOLO11n** (object detection, 80 classes) on 720p streams.
- Designed a 3-state temporal sequence detector for emergency SOS gestures with JNI callbacks; quantized models from FP32 to **INT8** via ONNX and TFLite, reducing model memory by 75% with negligible accuracy loss.

Cloud Infrastructure Performance & Scalability Benchmarking *May 2024 – Jul 2024*
Software Developer Intern, SmartThings Cloud Analytics, Samsung R&D Institute India – Bangalore

- Benchmarked **AWS EC2** vs. **AWS Lambda** across 8 cloud services to assess transactions-per-second (TPS) and cost-efficiency; configured Dockerized Node.js services with Auto Scaling and Load Balancers on EC2, and serverless API Gateway pipelines with CloudWatch telemetry.
- Simulated **70,000+ HTTPS requests** using **JMeter**; formulated architectural recommendations adopted for production cloud migration.

Academic Projects

Deepfake Video Detection via Cascaded Architectures *Jan 2025 – Apr 2025*

Course Project, EE655: Deep Learning & Computer Vision | Guide: Prof. Koteswar Jerripothula, IIT Kanpur

- Replaced heavy CNN-based face localization with Haar Cascade classifiers in a ResNeXt+LSTM video deepfake detection pipeline, accelerating inference speed by **up to** 5× for real-time frame processing on commodity hardware.
- Attained 76.5% **detection accuracy** on a diverse 2,000-video benchmark dataset while eliminating deep-learning dependencies from preprocessing.

In Situ Data Staging for Scientific Workflows | CS396A (UGP-II) *Aug 2024 – Nov 2024*

Supervisor: Prof. Preeti Malakar, IIT Kanpur

- Profiled the PQueue in-situ data staging library using Intel Optane non-volatile memory (NVM) in a WRF-based scientific simulation workflow.
- Measured fetch-and-merge, idle, and halo-exchange timings across 8W–2R, 16W–4R, and 32W–8R process configurations to examine performance and scaling trade-offs.

Python-to-x86 Optimizing Compiler *Jan 2024 – Apr 2024*

Course Project, CS335: Compiler Design | Guide: Prof. Swarnendu Biswas, IIT Kanpur

- Engineered an end-to-end optimizing compiler compiling a Python subset to x86 assembly using **Flex** (lexer), **Bison** (parser), and **C++**.
- Implemented AST-based 3-address-code intermediate representation, hierarchical symbol tables, static type checking, and runtime class support.

Topology-Aware Halo Exchange in Distributed Stencils *Jan 2024 – Apr 2024*

Course Project, CS633: Parallel Computing | Guide: Prof. Preeti Malakar, IIT Kanpur

- Implemented and evaluated MPI-based halo exchange algorithms for 5-point and 9-point stencils on IIT Kanpur's **HPC2010** cluster.
- Formulated topology-aware leader and non-leader communication schemes across 16 and 24 process grids to minimize inter-node data transfer.

Operating System Kernel Development (gemOS) *Aug 2023 – Nov 2023*

Course Project, CS330: Operating Systems | Guide: Prof. Debadatta Mishra, IIT Kanpur

- Engineered core kernel subsystems in C on x86 gemOS, including system call dispatch, function call tracing, and interrupt handlers.
- Implemented virtual memory address space management (`mmap`, `munmap`, `mprotect`) and Copy-on-Write (CoW) process fork semantics.

Relevant Courses

Deep Learning & Computer Vision	Operating Systems	Data Structures & Algorithms
Introduction to Machine Learning	Computer Networks	Design & Analysis of Algorithms
Parallel Computing (HPC)	Compiler Design	Principles of Database Systems
Computer Organization	Software Development & Operations	Discrete Mathematics for CS
Probability for Computer Science	Logic for Computer Science	Linear Algebra & Differential Equations

Technical Skills

Languages	Python, C, C++, Rust, SQL, JavaScript, Bash, Verilog HDL, MIPS Assembly
ML / AI	PyTorch, TensorFlow, TFLite, ONNX Runtime, Hugging Face, OpenCV, CLIP, MobileCLIP, YOLO, ResNet, NNStreamer
Systems & Tools	Linux/Unix, Git, Docker, SQLite, AWS (EC2, Lambda, CloudWatch), MPI, GStreamer, JMeter, Flex, Bison, LaTeX
Libraries	NumPy, Pandas, Matplotlib, Scikit-learn, React.js, Node.js

Extracurricular Activities

- **Inter IIT Sports Meet**: Goalkeeper for IIT Kanpur Hockey Team at the 55th Inter IIT Sports Meet. *2022*
- **Senior Executive, Udghosh '22 (Hockey)**: Coordinated operations and logistics for 16 matches. *2022*
- **Inter-Hostel Sports (Inferno '22)**: Starting Goalkeeper for Hall-12 Hockey Team, securing **1st position**. *2022*
- **Cultural Activities**: Represented batch in competitive Group Dance at Freshers Showcase, IIT Kanpur. *2021*
- **National Cadet Corps (NCC)**: Completed 10-day Combined Annual Training Camp (CATC) at Rajpipla. *2017*